

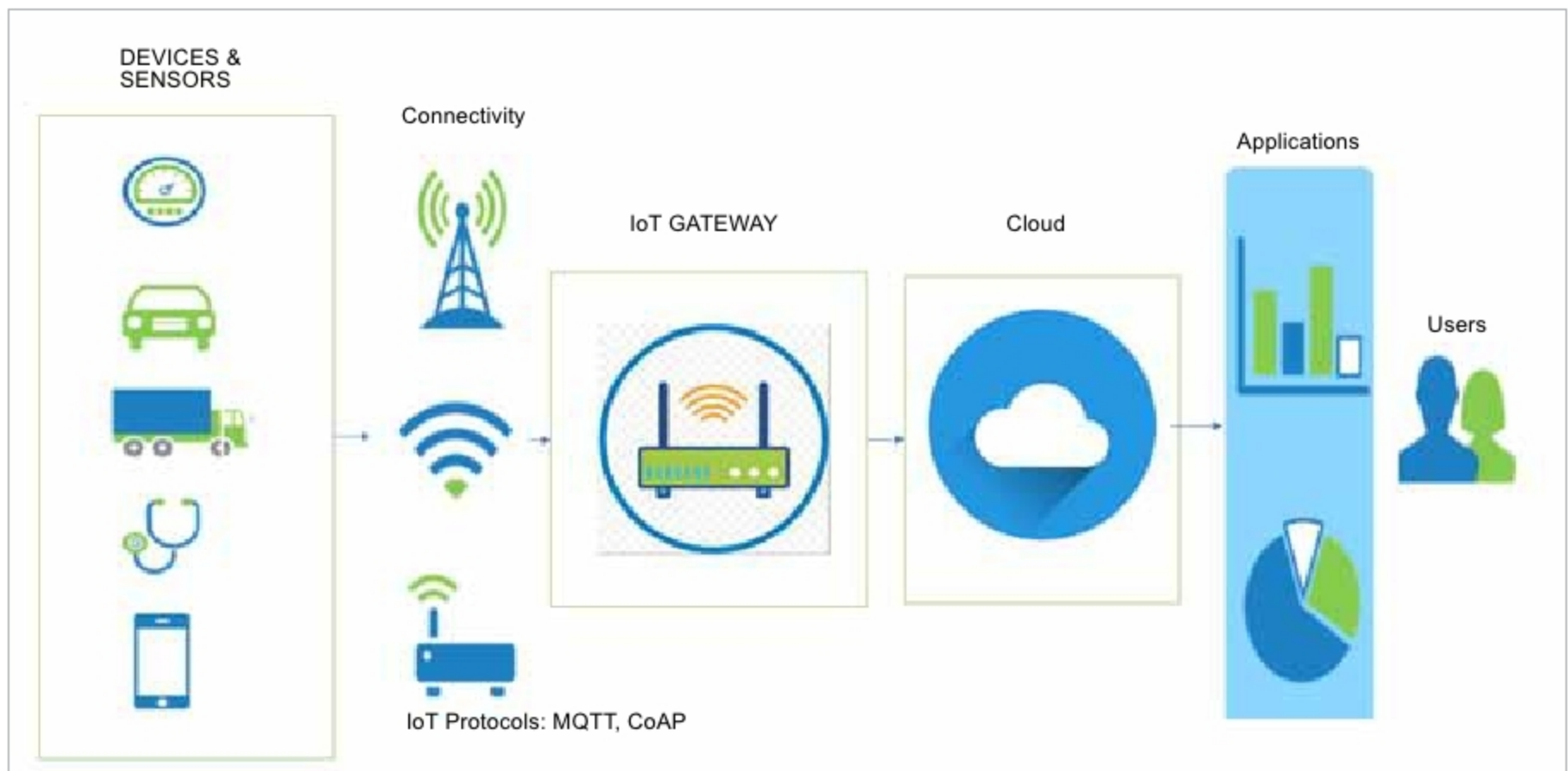


Figure 1: IoT architecture

When the public cloud is used as the data management and analysis platform for an IoT application, the major concerns are the latency involved and the cloud connectivity intermittence, especially for mission-critical and real-time applications like connected vehicles or other autonomous systems. Keeping sensitive data in the cloud is also not something many enterprises want to do. Further, when large volumes of data need to be stored in the cloud, data migration becomes costly.

Edge computing architecture

Edge computing solves the above limitations by bringing storage, analytics and intelligence close to the data source. It allows IT to retain sensitive information on-premise, while still taking advantage of the elasticity offered by the public cloud. The cluster of edge computing devices, which handles network switching, routing, load balancing, security and the audit trail, becomes the local ingestion point for data originating at various sources.

Figure 2 shows the 3-tier edge computing architecture. The first tier is a data layer that includes everything

that is a data source. The second tier is the intelligence layer that spans across the public cloud and the edge layers, wherein the cloud layer does the heavy lifting and the edge layer deals with the production and real-time data sets. The third tier is the insights layer that presents accurate insights delivered by the intelligence layer to the users through rich dashboards.

Fog computing adds a layer between the edge and the cloud. It moves the computing activities to the processors that are connected to the LAN hardware (where the IoT gateway is), which is more distant from the edge devices like the sensors and actuators. Fog computing has the advantage of being able to connect to more devices and process more data; however, it requires more investment than edge computing.

IoT protocols and edge

A few transport-layer protocols that are widely used to form IoT communication links between the devices and the IoT gateway are Bluetooth, Wi-Fi, NFC (near field communication), cellular, Ethernet,

Zigbee, LoRaWAN, WiMax, etc, to name a few. They differ in their range, ease of setting up, cost and bandwidth.

IoT data protocols define the syntax, semantics, synchronisation of data, and error recovery. A few widely used IoT data protocols are Message Queue Telemetry Transport (MQTT), Advanced Message Queue Protocol (AMQP), Constrained Application Protocol (CoAP), RESTful HTTP, Mosquitto (an Open Source MQTT broker), Extensible Messaging and Presence Protocol (XMPP), Data Distribution Service (DDS) for real-time systems, Lightweight Local Automation Protocol (LLAP), Lightweight M2M (LWM2M) and Simple Sensor Interface (SSI), to name a few.

Components of edge and the open source technology stack

The edge gateway (or the IoT gateway) processes the data collected from devices and sensors, and converts it into expected formats. It establishes a bi-directional connection between the IoT devices and the storage and analytics system. It protects information